

Intel ipu6 otocam driver User Manual

Revision	Description	Release Date
V0.01.00	Initial Release	May, 19. 2026

1. Get started Guide:

- While setup the build environment on the new ipu6 machine, do items from here:
 - follow the steps in the doc “828853_ArrowLakeUH_Ubuntu_Kernel_Overlay_GSG_3.1.pdf” and fix the steps of session 2.3.2 for building the linux kernel:
 - follow the steps:
 - (a) `sudo git clone https://github.com/intel/linux-kernel-overlay.git`
 - (b) `cd linux-kernel-overlay`
 - (c) `sudo git checkout lts-overlay-v6.12.61-ubuntu-251229T065211Z`
 - (d) `sudo ./build.sh -r no -t lts-overlay-v6.12.61-ubuntu-251229T065211Z -b 1000 -c lts`

2. Driver installation :

- (a) file: `$/etc/modprobe.d/ipu.conf`
add one line: `options intel-ipu6 isys_freq_override=475`
- (b) retrieve the otocam-intel-driver:
 - `$cd /home/user/otocam-intel-driver-vx.xx.xx/script/`
 - `$chmod 777 *.sh`
 - `$/install_image.sh`
 - `$/install_deb.sh`
 - `$/install_script.sh`
- While update new ko or xml, do items from here:
 - (c) copy the ko & xml to the intel system:
 - `$cd /home/user/intel-driver-release/otocam-intel-driver-vx.xx.xx/script/`
 - `$/install_script.sh`

3.Steps to execute sensor preview:

- while change new camera, do items from here:
 - (1) reboot the intel machine
 - enter BIOS
 - modify the HID for the specific sensor :
 UEFI Firmware Setting ->enter BIOS...-> Intel advanced menu->System Agent (SA)
 Configuration -> MIPI Camera Configuration -> Camera1&2 Link options
 - ex: `otocam222 => OTOCM222`
`otocam223 => OTOCM223`
`otocam260 => OTOCM260`
`otocam271 => OTOCM271`
`otocam276 => OTOCM276`

BIOS Setting is based on "863425 IPU6 GMSL Enabling Guide Rev1.0.pdf" and fixed as following:

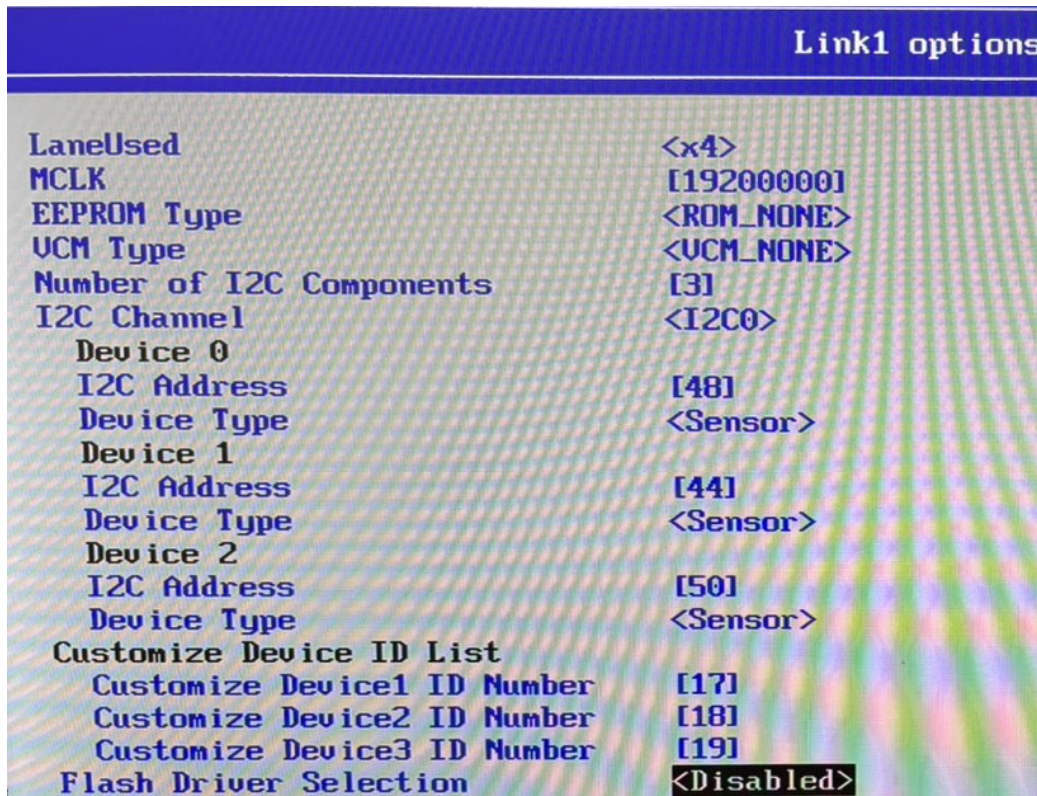
- (a) PPR Value = 2 =>2 lanes
4 =>4 lanes
- (b) MIPI port = 0 =>link 0
port = 4 =>link 1

MIPI Camera Configuration		
CUF/CUS Support	<Disabled>	Camera1
Control Logic 1	<Disabled>	
Control Logic 2	<Disabled>	
Control Logic 3	<Disabled>	
Control Logic 4	<Disabled>	
Camera1	<Enabled>	
▶ Link options		
Camera2	<Enabled>	
▶ Link options		
Camera3	<Disabled>	
Camera4	<Disabled>	
Use GPP_F6 as HDMI_WAKE	<Disabled>	

Link0 options	
Camera1	
Sensor Model	<User Custom>
Custom HID	OTOCM222
Lanes Clock division	<4 4 2 2>
CRD Version	<CRD-D>
GPIO control	<No Control Logic>
Camera position	<Back>
Flash Support	<Driver default>
Privacy LED	<Driver default>
Rotation	<90>
PPR Value	[4]
PPR Unit	[2]
Camera module name	09B13
MIPI port	[0]
LaneUsed	<x4>
MCLK	[19200000]
EEPROM Type	<ROM_NONE>
UCM Type	<UCM_NONE>
Number of I2C Components	[3]
I2C Channel	<I2C1>

Link0 options	
LaneUsed	<x4>
MCLK	[19200000]
EEPROM Type	<ROM_NONE>
UCM Type	<UCM_NONE>
Number of I2C Components	[3]
I2C Channel	<I2C1>
Device 0	
I2C Address	[48]
Device Type	<Sensor>
Device 1	
I2C Address	[44]
Device Type	<Sensor>
Device 2	
I2C Address	[50]
Device Type	<Sensor>
Customize Device ID List	
Customize Device1 ID Number	[17]
Customize Device2 ID Number	[18]
Customize Device3 ID Number	[19]
Flash Driver Selection	<Disabled>

Link1 options	
Camera2	
Sensor Model	<User Custom>
Custom HID	OTOCM222
Lanes Clock division	<4 4 2 2>
CRD Version	<CRD-D>
GPIO control	<No Control Logic>
Camera position	<Back>
Flash Support	<Driver default>
Privacy LED	<Driver default>
Rotation	<90>
Voltage Rail	<3 voltage rail>
PPR Value	[4]
PPR Unit	[2]
Camera module name	A12N08B
MIPI port	[4]
LaneUsed	<x4>
MCLK	[19200000]
EEPROM Type	<ROM_NONE>
UCM Type	<UCM_NONE>
Number of I2C Components	[3]



(2) enter ubuntu to preview image. (Do it every time)

- \$cd /home/user/otocam-intel-driver-vx.xx.xx/script/
- \$sudo su
- \$source env.sh
- \$./otocamxxx-xxx.sh (ctrl-c to stop steaming)